

Introduction (Lecture) ~ Avel Guénin-Carlut ~ Active Inference for the Social Sciences 2023

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all right hello and welcome everyone it is July 7th 2023 and we are in the first introductory session of the course active inference for the social sciences this is going to be a really exciting course and journey that we're all going to join and co-create together thank you complex systems and feedbacks with cyber physical systems will be a theme we told you it would be I will pass now to Avil for the introductory session if you're watching live please feel free to write any comments or questions in the chat and I'll relay them at the end on to the lecture thanks Evel go for it foreign okay cool so I'm available I uh organized against the course I uh had the ID and uh worked on the curriculum uh do you see this slides well looks good and just make sure to close the YouTube video or mute it so hi uh some issue okay cool okay I did not know how to do that so uh this is the introduction session of the course Very unsurprisingly my name is Regina kerou I am a founding member of careless research and I am a non-funding member of Q inference lab which uh co-organize the course and first I will basically present the teaching team so we have Danielle which is uh with us which is a PSG ecology and evolution studying and we have your in Eco Evo kind of way who is a president and co-founder of the activator Institutes and who is interested in Hamilton's art philosophy and cognitive security then we have done which is also here with us which is a second year prg candidate in philosophical University of Sussex which like me interested in um for e-cognition which is um embodied inactive embedded and extended cognition we don't need to get into that now he will explain to you and in predictive processing and which is basically working on the cognitive science of technology and then we are the people who are not here today which are Mao Al Bahasa tsg candidate in the cognitive Computing at University which is head of product Innovation and research at versus private sector research lab in Ai and the mathematics of AI which specializes in Wildlife inference so we were working on right now script Theory which you will extend maybe I will talk a bit of it today but she will be the one

who explains what Chris criteria is to you and modular activity artificial intelligence and we have Learning Center also appearances also in philosophy but in Wollongong I believe this is how you say it's in Australia which is centered in Philadelphia mind and in the status of information processing as the term is used in cognitive science and then there is me so I made a career timeline because when I talk about what I did people get confused I did a physics math I did engineering I did cognitive science I did research in ecology and evolution and in the Masters in a complex system physics and uh I did like one year apparently given my timeline or two no more than more to us independent research as a part of keros research and as part of the French society of Economics before I started the phdn ring now which is a something about the relation between mind and material culture we will get into that at some point and the question motivation I have are um essentially describing the relation between phenomenology so the way we're experience the world and the physical reality which I have The Fringe view that somehow of criminology is constructive of physical reality which could have weak and some interpretation I subscribe to both I'm trying maybe more importantly here to uh explain or at least describe how Social Security life format to say the irregularities that analyze social life humans through life are constructed which is hard and I'm trying to uh well build a research that is effective at doing you know knowledge not word piles through our work in chaos research so these are specific reference you will get the slides at some point I guess I will publish them on the website and then we'll be able to check it out if this is of interest to you so let us get on the actual relevant content so the what we are going for here is basically explain the very basic the merge of the course explain um how it's of relevance how the framework we proposes of free events to social science and present a few case studies you know make the case and then we will talk about course layout but this is pretty quick first I'd like to talk about the explanation of cultural Dynamics what does explaining cultural Dynamics even mean it depends who talks there are many many many many many types or views on what is scientific explanation I say we can cluster them in three broad categories one is a numerological so it's about laws and what it does is to take regularities just say here is this regularity and consider that these statements so the law has an explanatory capacity toward the regularity so for example for those of you who are familiar with um the work of and I forget names Foundation the foundations Trilogy this is pretty close to what they call Psycho history so basically trying to do trying to find laws of behavior and using them to predict history in a regular lawful way then we have mechanical explanation which are more typical of the life science and what we do in mechanical explanation is to look at the structure

of a system and model the structure and try to use this model to you know predict specific regularities that are of Interest and you have a approach to social sense that is I'd say quite close to that which is cloud Linux logical side but still is centered on Dynamics and mechanisms and explanation by calling on to structure in specific societies and then you have a functional explanation which asks what constraints exists so that the system does what it is understand those here again you will also use a lot of contextual modeling very often evolutionary Game Theory so simulation where we suppose that agent can have specific traits and those that reproduce more or that have more Fitness you know give their threats around and you have the discipline field of study of cultural Evolution that does that very concretely by just asking you know how could you evolve so what cultural threats can get more Fitness and be transmitted a very evolutionary centered approach to social change so uh to send all of this up basically we explained things by calling on to regularities always and we can't talk about regularities in the phenomenology so what the system does in of itself in this structure that is no language system activity or in the function that is the constraints that exist over what the Sim does and you have a specific brand of explanatory strategies that try to legify behavior to make nomological lawful explanation of human behavior and of such a cultural change and it's not immediately evident how they do that how they think it's coherent ideally to begin with because well humans learn cultural challenge so what laws could they be in um in a field where which is changing with time always in my opinion it is a more question of you know rhetorics than of epistemology because I do not think the answer is coherent the answer is as far as I'm concerned that people they just refi regularities that are observed in experimental settings often that are quite heavily fabricated as kind of cognitive laws of behavior and because those people usually agree in spirit to the mechanical kind of explanation they talk about cognitive mechanism or mental processes but only insofar that those mechanisms or processes are explanatory of cognitive laws they do not do the work of you know looking at the structure and function of the systems that I just claim there must be a mechanism that explains the vague law that I have observed over 20 people that came to my lab and these laws they are assumed to scale for the inclination of social activity so you have it you have the recipe for universal law of social change and the basically the whole arguments stands on the idea that you have a presentation or mental processes or a cognitive mechanism that somehow entail the subjects behavior is regular through time and between individuals and you never have the work that would actually in any capacity prove it for reasons that we will not get into because it will get math ready and renty if uh the rules and regularities of human behavior can

change to learning through cultural Evolution through whatnot how can we exactly model or explain or even describe this agency um this is not doable in an homological framework because when we do natural laws we look at regularities we claim here is alone this regularity is a law so it explains that you know it exists and uh this is a circle because as far as I'm concerned it's a check log reasoning like the regularities explain the regularities which explain the regularities there is no way out of this you can recurse this all day long this the only fact that you observe the thing is restoration for the thing this is not hopefully how we do things in science any other process of explanation that is more complete if you look at uh structure and function written large you can look at a specific system which has specific structure and you can look at what the system does well the function of the system and at the specific time and you can look out at how this function this activity shapes the system later on at t plus one you can have uh you know developmental approach to things that is internal health system unfold how they learn how they evolve through time like that is doable that there is no law against that and it could actually work so yeah let's do that um let's explain natural Fernand by describing how the underlying structure work but also by trying to explain how it's brought about how resistance activity explain the way it's takes shape through time so a claim that we will make here that is itself it could be contentious but we don't have to look at why it is not Philippines here is that active in France which is a specific new trendy theory of human cognition can actually explain how regularities in relying social activity be constructed Through Time and let us look at what active Insurance says precisely so there is we won't look at the math ever we won't look here at the specifics of the theory the only thing we are going for is to prove that this Theory indeed does what I think it does which is explained or at least described in a non-trivial way the evolution of you know social regularities if active inference framework is true then cognitive agents perceive the world through expectation they have developed about the world so perception is an active process I do not perceive things with my eyes and compute higher and higher level of traits of what I saw I basically expect things in front of me and I will actively look at the things that are most likely to inform me and something that is critical about this framework is that the prediction it also underlies action or policies so let's say trajectory of action that inact I will do the action that I believe I'm more likely to do I will do that in virtue of me believing I am more likely to do this action oops in that framework active in France you understood as a process of an action of a specific model that is causally shaped both by an agents Dynamics and environment Dynamics and this relation well brings a model that corresponds in abstract way to the um structure of the

Dynamics of the relation and this module I can also look at um look at it in a way more concrete concrete sense in that it is what in the lives what drives the edge and dynamic you have a duality between the Dynamics and the epistemic mantle states that we put that we project that we can observe in the learning a very important aspect of this is that it dissolves entirely agency as we would understand the term there is no distinct step of perception which informs the agent and what action on what the world is like and based on which the agents decide freely what action it will take there is a constant prediction of what is going to happen and this is enacted by the agent both through perception and action so the what drives decision here what drives a generally the flow that drive that analyze the Urgent behavior is the Urgent self model and that is that has interesting consequences so I have notifications so we'll look at them sorry what let me drink I will drink I need to drink so one consequence of this um formalism is that normativity so the values and Norms it's hard to go around that word that drive Behavior they are not you know in the brain they are not something that the Urgent abstractly relates to they are embedded in the very process in which the agent understand and gojis and navigates the world and this is a variable event for a human cultural Evolution because well we are cultural creatures we learn expectation about each other uh both explicitly by you know looking at how people do things and then expecting them to do the same things again and implicitly through the very process by which we relates to other people like Norms like language and the way you look at this this um this learning three Mission it can be it can be reduced essentially because of the properties of active infronts to uh what people call regimes of attention so the way I pay attention to things the specific things I will look at in my process of understanding and navigating the world the Flow by which admind that is the same thing as a set of value and a set of expectations what is important here is that normativity and values and things like that they are uh embedded in my cognition in a much more basic way than you know conscious self-reflection also of course conscious reflection can inform that flow it is uh embedded in the shared expectation I have with other people and there is no Valentino result that I can actually be reflexive about it because well the web predicts things it's based on the statistical regularities that have seen in the world and because the way I pay attention it's you know entailed by a proactive model of what is going on around me I'm very constrained very much constrained by the statistics I have of the thing I have seen around me so regularities Drive Hawaii pay attention which Drive implicitly or specifically value and normativity and you cannot demical steps you can add reflexivity you can add I don't do accountability you can add a lot of things but you cannot remove the very basic way in which

cultural values and transportation are embedded in perception so um yes this is a repetition of what I said the patterns in the flow of attention that are embedded that I have developed through the very fact of engaging with a cultural niche and they constrain my actions towards specific outcome they entail specific values specific norms and they entail um at least they constrain very very very very very heavily the intention that I'm even able to conceive to generate there is a dual Liberation equivalent way to see this which is uh engagement with the social landscape so every agent they experience the world not in the same way of course but the experience the same let us say that each person this kind of the world it's predicts very strongly the experience of the world that other people we develop around them so they are structural properties of the you know landscape of opportunities and perception and possible action that we experience there are properties that are transferred between individuals robustly this is the case for example of norms so if I say if I say um if I look at the law any specific law all of the people in the same country they are subjected to the same law so maybe some of them can just ignore it they live too far from the city we don't have cops here maybe some of them are above the law in some capacity of their relation but the basic constraints that the law entire it is experienced by everyone in a different way we all expect that specific people will react to specific situations in specific way and this is a special property of the landscape that we experience the way that these landscape shapes expectation in a robust way we can look at basically it's as a constraint so not as a iterative property of a landscape that I can or cannot experience but just as a statistical force that shapes behavior in a specific way and looking at things like that it means that I can see uh given social cultural feel in two complementary way as a set of agents that navigates uh common well not quite common but mutually constrained social control worlds or as a set of constraints that produce itself and the way agents reconstructs the world around them in their mind the way they understand it you know you know creative manner it is equivalent to the way the set of constraints reconstructs itself which is important because if I can reconstruct if I can look at this real world as a set of constraints over statistical constraints for behavior that reproduce itself I can look at it as a living thing strictly speaking because this is how we Define living things at least in specific sub-branch of system biology that I believe is relevant is to be believed about that so when we talk about structural landscape we can look at them either as a landscape of affordances this is opportunity for Action that are experienced and enacted by a set of individual agents which interacts so as to having some level of coherence between the Landscapes they um experience Linux and I can look at the same landscape as a system of constraints that are exerted over the

behavior of these agents and that I can define and study with no play reference to any individual agent in the same way that I can study a I don't know dog without looking at every single individual cell what you're interested in biology or sociology is usually you know aggregated patterns not what every single living thing around thinks of every single thing that was the heaviest it got I uh probably over the world course I do not expect you to grasp all of the consequences of what I said because it was very condensed and it was very abstract but the case I'm making here is not that everyone has to understand all the math right now it is uh that's this framework it does things to say about relevant question in social science and to make this point I will not just you know give the abstract presentation of how it's awesome because we have landscape what not I will look at specific fields questions case studies in social science and I will look at basically how actually in France entails a pretty specific model of that let's see as well grounded in commutative sense and that is relevant and that is adequate to the study of living changing systems etc etc first I join us a team some of you will know what it is some of you will not know what it is address I think is something that is situated in the work on public opinion specifically like we I believe this book is called public opinion um so another setting it refers historically also you have more diverse way to understand it now to the role of mass media like journals and later TV in manufacturing consent this is a textual station from the text of um no I thought in the book cover I forgot the name of the guy I'm sorry you have to look at the reference but it is used to study how mass media can manifest manufacture consent by drawing attention to specific issues and shaping specific expectations about the world so the idea is that you have a role an active role of the elites what they call the elites here people which have outsized influence of the flow of policy or society I will call them elite and so those people they basically transmit norms like transmit non-native expectation uh so consent to whatever is the thing that they do that they want you to do more specifically and I do so not by telling you uh do this and that they do so by building agendas so drawing attention to specific issues in specific way and you can see here people who are I mean it's prevalent in political like horrendous that but the U.S right wing is very very very um transparent in doing that and so this is a meeting between two experts in Azure setting with Donald Trump and run the sentis which basically I've convinced people that they should not worry about things like that I don't know the price of life whether we wage war here and there uh why do we have to pay millions why do U.S people have to pay millions to get healthcare but to how LGBTQ democratic people threaten the US life in a very abstract way like a did not ever hear any consistent coherent believable mechanism by which they explained how LGBTQ

threatens whatever Republicans uh but they do draw attention to this topic they do share this rotation that Democrats and gbtq are collectors and they do so in a way that is able to legitimize themselves as become other people to mobilize people to actually you know be behind them and do their bidding and to colonize political life in a way that is convenient for them because it you know keeps any power and the role of attention uh the way you pay attention to things in shaping Behavior it's just a redo of what I explained in um in active inference if you believe that the way we the force is the way we constrain attention is applicative of behavior then you believe in actual inference just Express differently so this is a case where you have a classical let us say concept industrial sense that essentially reduces to concepts of active inference it's not to say of course that sociology is a relevant and we should just do cognitive sense but this is pretty rare to have such a direct response between topics that are disciplinary this far apart so all this is something we should definitely pay attention to so uh second case study nationalism um basically this case study is entailed by what I said over the Republicans because we move by which they legitimize their rule it is just say we have we share an identity here um U.S conservative Traditional Values etc etc and this identity it entails specific norms and for reasons that we originally left out of the frame I so Elite people run the scientist Trump Napoleon or whatever I get to tell you what this energy entails in terms of behavior you don't get to speak really because you know you are not someone who has a newspapers that para today say you know what you say because you have to keep on doing this subject and this is very consistent with Dynamics by which we constructed nation states so for the uh historical context the idea that we have ethnicities and this is somehow relevant normatively like this love should organize together or the uh I don't know uh hungarians together and we should get to agree to things that we do and then do them it's very recent um it's extremely recent dates well I won't get into the specifics but let us say in this specific formulation it dates from the 19th century uh when you add a bunch of very poor photographic Administration that took hold respectively of France United Kingdom uh crucial and they are the capability to rule most of these social activities but they did not have any clear argument why people should you know listen to them in any capacity and the way it um it worked is that those Administration they basically started claiming that there were some representative of a given uh ethnical and linguistic community and that in virtue of being part of this community you should listen to them and uh pay them respect and actively work for their goal to be fulfilled and of course it is not that easy to make this claim but what happened is invested a huge deal of inner generalization into actually having people buy that they built

schools a little prison they built a psychiatric hospital they built a lot of infrastructure that enabled them to transmit Norms in a very robust way and transmit identity in a very robust way and in doing that they managed to embed most of the micro relations you could have into their specific ethnicity and into the normative power they lend to this ethnicity so at some point you were you know um I always forget farmer in uh lost Village in France you only meets menstruation at tax time so main arm come and tell you give your grain and you give your brain because that's your dad and at some point those people send you a teacher and they send you newspaper and they claim hey you are French you have to speak French you have to be friends with us you have to fight our Wars and because there is so much you know uh energy that is pending that people actually bite or die that's another option because you know uh the uh other side of the coin is that states that legitimate themselves through a nationhood they are basically to be all the time invested in at Nicole cleansing because people who are not the right ethnicity they are not considered element of these states so they have to be submitted or removed which is a bad thing to do this is very bad thing to do when you can see um I like I'd like to illustrate this whole notion of energy capture through the African Renaissance monument that Abdullah wait uh so the president of Senegal had constructed in uh in the capital so it's very glorious it is very um good example of identity capture because it shows you uh the Norms it shows your glorious image of the African man and it shows how you have to you know be supportive of children show them protect them with your big muscle and give them very well ambition while African women look at you like they were non-agents let us say that so it gives a blueprint around which society should be organized it gives you a very specific picture of who this is speaking to in this case Africans because I guess Senegal gets to say what is good for Africa that is um coherent claim to make I mean sarcastic sorry and why I like this specific Monumental you have lots lots well first it's very expressive you don't you're the auditory of in France it serves the basics and function but it's not it does not show you it tells you and the afghanistani shows you what it means in a very uh well obvious way and the second main reason is this uh you have nothing around it they just put it in a pile of dirt and I invest in nothing in the infrastructure because they were too busy you know keeping taking money from themselves I guess I don't know exactly what who I did with tax money but clearly it was more invested in the symbolism of what African men should do and why they should follow him specifically and not in concrete infrastructure and in most cases energy capture is very radial for people who are you know captured this is why I call that capture because uh what you get out of energy capture is a symbolic sense of being part of the good

team and what you have to give is usually you know the entirety of the work you do over your life uh it gets given to the state Administration that wants you to do things and yeah you do things for them and sometimes they build infrastructure but if they don't want to they will not that is very very very very much a bad deal and I uh I give it away so National entity it was diffused to a school and mass media it was embedded in a micro networks of solidarity so typically the African resource Monument it gives you basically tells you that as a huge bond in fatter you have to give in whatever the state analogy of what uh was and then it is leverage for mobilization and legitimation and control and you have another that's that's not um oh I did not say why it was relevant to a team friends well it is relevant to active in France because all this is mediated by self model all of this is enabled because people have self model and they enact the self model and there is no necessity of a reflexive step where you actually you know think about what is yourself model and oh you are expected to do a thing you do the thing because you expect yourself to be doing the thing and relaxing in France as we said you do what you expect to do so this basically explains why such symbolic you know raw power it has actual consequences well if you have uh you know more rational model of humanity the question is hey why you build big things and not uh infrastructure it would be a rational response but this is not what people do people follow norms and this is a way to broadcast Norms very efficiently and robustly so uh due to that I'd like to talk about structural agency so there is a constant question in a social sense usually social sense talk about you know structure so Norms habitus uh way that institution builds through life and there is a legitimate question that is asked which is if you believe structural treatment Behavior what is the place for agency so there is there are constant debates in every field possible of social sense regarding what is actually explained by the structure and what is left to individual agency and what I want to say is that this question is especially dissolved by the model of agency we have as we said uh agencies exerted through expectation over myself so because social structure it becomes expected it is enacted through the LMC of these specific people that are within it not have an opposition between how strong the institutions are and how adjunctive people are because strong institutional institutions that basically become embedded in yourself model and so are um you know there are people freely not freely it's a matter but from their own accord they enactose institution and this is the whole thing about anti-capture which I presented nationalistic ideology is strong is efficient in building actual mobilization because it makes you believe that you will actually if you want a court follow the uh um social order it gives which will which was often presented but pretty rare in a pre-modern institution usually from

the institution are um I am a king I have an army give me food and when you have legitimation attempts to actually explain why I should get the um these are pretty much justification this is the elite people talking to themselves about what why the objectified knowing that but you have no arguments and you have that would be actually compelling so as to why you would have to give the food and you have no attempt whatsoever to uh convince people who have to give the food they should give the food the psychology is not they do not talk to the people they ask things from they talk to themselves about why they have they can't they are legitimate to take those things and so yeah in the in those uh case you have a pretty evident controlling between structure and agency do I uh do what I'm told to do on their set of death or do I do any other thing of course most people would like to do other thing but they can't and I will feel or they can't but if you accept that agency is built through normativity through engagement with a social cultural field of finances Authority for action yeah you have to admit that software exists at least in part through agency because what agent believe is legitimate what agents believe is right what I don't believe is expected it is very precisely what considered sutures and that is a Insight that is as at least in partly in of the conversation in in some killer I'd like to um to show why it's um the direct conflict between certain agencies it does not have a strong running uh uh yes again I have said the thing that the slide would say there is another thing that is maybe closer to my core body of work which is uh how active in France afford indirectly an archeology of an interview landscape if such organization is made of constraint or behavior and those constraints are to some extent embedded in the natural environment of the people then we count to some extent study archaeological landscape for themselves we can look at the landscape like material landscape the um objective study of archeology and we can make some inference over the constraints that this landscape entails of our Collective Behavior uh I will not get into detail so as to why because it's um it's uh Farkle making this inferences difficult but very concretely we can ask how whether you're okay so daddy that some people get to say to other people what to do and certification really that some people are by nature or both other people uh there can be a very vertical patterning so you have a claim by um people who fund me escape projects look it up it's nice you have a claim that to some extent the rectification of society around the Iron Age it is written it is embedded in vertical patterns in material culture and architecture and maybe I will not buy the claim outright but it is very compelling if you look at specific architecture you get a specific feeling so this is the um not cathedral but maybe basilic or free catic I will not try to say the name because I do not speak Icelandic and that would be bad but you are very Progressive

compelling curve throughout the sky it was a specific feelings that are um how do you say produced by this architecture and you have a very specific way in which drives attention from some something wide on the ground to something that goes up and if I did not make it clear yet if we buy it in France we buy that the patenting of attention uh the way you give attention to things it carries Norms it carries intentionality and in first person by introspection you can I think all of you can very clearly see how architecture it drives specific patterns of attention and how those pattern specific elicit specific feelings and we currently do not have models that would be very specific and reproductive of how this work but we could this is this is usual but this is doable work this is not speculations and I do not know further communism but actually in France that would actually make it possible and last I'd like to uh talk about uh the world's work my favorite thing in Social sense I guess which is a single state Scots anthropology of states so the core argument of Scots I'd say it's hard to identify a core argument in that book but you have ways in which states so here equivalent to administrations they understand the world and some world through you know written forms then they can pass around uh many people one institution to have actually a coherent model of the world but he's that is pretty much obvious and what should be obvious but is not is that those models are reductive in the sense any model are they emphasize specific information and they live out other specific information and something that is very interesting I'd say is that states they tend to push specific models specific Landscaping specific Norms in urbanism specific family norms the build eligibility so the ability to be understood by the state so for example uh you could have you could have very sparse settlements that are quite autonomous and that do the thing it is a possibilities yeah there is no law or physical constraint to run that but it is far easier as a state to manage what is very close to it right so for a state it's good to have you know a population that lives very close to a center of power because those people they are just there you can go tell them what to do it's simple and you can you can go ask them things it's simple too it's very easy to manage and a lot of people they would need you know food to stay there so we need some kind of food production model that is easy to take from that is easy to organize and engulge as a centralized Institution and this is monoculture this is agriculture as we understand it currently this is people I think Big Field that make the same crop always cereals well not always sometimes potatoes but mostly cereals because Heroes you know it's very dense energetically it is easy to keep it's easy to transport because it is dense and it is it's Day Edible long after you know recalled user recalls English crop cropping I don't know what is the word but when you take the food from the plants and people have to stay there for very long to

cultivate that so they cannot you know go around and put potatoes here and there and leave when they don't so you can just go to people and you know where the people are and where you can take the food and uh you know when the food is going to be um to be again I don't know the word cropped because materials it's always at the same time it's very visible in the landscape uh how far you are from uh resulting slash cropping so you cannot really have a way to cheat that you can just send many nerves take the food give it to the city to Japan population and you have a very um rationalized controllable Society we we do not know that this is the reason why we have cities and uh monoculture but we do have traces that states very consistently try to have big cities and more cultural field and mostly they succeeded and oh yeah this is relevant to active in France because inactive in France the way you pay attention to things it is productive of your world it is productive of what cognition you have and here the capacity for administration the capacity for written symbols that tell what things are they are critical to States understanding of the world they are what literally produces the way they States pay attention to things and in fact it wasn't basically major thing for them to have a world that looks like this it is uh very directly telling about you know the fact it creates a world it's a direct institution of that and that is a nice thing to do and I'm happy I did this link so this is for the case studies uh uh expectedly you don't see the collections not all of them maybe you're convinced that there is something there but most likely you don't see very precisely what is to be seen here so the expectation is that over the course you develop a very specific understanding of active inference the framework uh you develop a specific understanding of how it relates to social science to questions that are relevant instruments and we develop the capacity to know of a informal opinion that's critical inform opinion on how this relates to question in true science uh we are doing the introduction lecture hopefully I have convinced you there is something to see here if not too bad uh and the next session will be in four days not one week four days it will be about um basically the basics of action France what you have to know to understand not the framework per C because it's math heavy but what the framework says about all life as cognitive agents so then you have the discussion I you should know that for each lecture there is discussion lecture is a you know vertical people told you listen discussion is more free form uh every person has a magically Authority on how they share the discussion but the expectation is that it is a discussion because it's the name we gave to the thing so after the basics we'll look at patterns of collective Behavior so how those elements on the cognition human combination are non-efficient they give tools to study the organization of collective systems like and colonies our societies but we'll stay in um

not specific to human approach we'll be interested in Collective Behavior but we're not interested in the specific patterns of collective Behavior as afforded by a human condition they will look at semantics and semantics the evolution of symbols so things that are closer to humans we look at how at in France accounts for evolution of symbol and what evolution of symbol affords for commission and then from here we look at various efficiency things Norm script narrative language so how thesis cushion on active inference and Collective behavior and semiotics they inform very directly they give a model specific principled predictive model of social human behavior and then we look at so my thing the way I like to think of it all which is show constraints and we'll go back to the claim that their Landscapes of social cultural commission thingy thing and those landscape can be looked at in uh you know abstracted from the individual scale Manner and this is explanatory of control change and then we go for the conclusion so we're interested in your opinion of course this is the first one we'd like to have feedback a lot of feedback on what you thought of it and what would make him good so that will be for the conclusion discussion so here I hoped that you would present yourself uh we have not the capacity right now to have um students tag the screen and talk but I'd like to know uh you can say in the chats what is your background what's our expectation for the course whatever you want to say about you know things the world preferably um being related somehow to the course that would be nice uh but yeah please tell things to us that concludes my talk I thought these slides but clear we have uh last thing maybe we considered making uh projects so you making projects that we read and maybe on Thursday grades but I do not think that we are in capacity to give you know um and standardized normalized degree or at a station that has any value so we decided not to do it but if you so asking you to do things uh to do projects is a bit weird if you're not gonna you know grade it and attest that you did it but if you want to do it if you want to write things and send them to us for review this is a this is very okay you can do you can do that you are not formally invited to do that but you can do it well I'll just add a point on that and then meanwhile if anybody wants to ask a question live chat we did raise this topic of different qualifications and it's something that we've been working on and continue to work on in The Institute so potentially in future versions but absolutely in future offerings we will have these kinds of courses but as eval mentioned this is kind of a first run and so it's really exciting just to even start that conversation and and see what people are looking for in terms of certification um all right so let's go to some questions in the live chat and comments oh yeah good um I did not make this escalation clear but you know because it is a first and because we are still checking the course

it is so much more important than you actually tell us what what you expect out of it like what you want or what you perceive will be here you know expecting the active influence sense that is a bit perception a bit action please tell us about that awesome in fact the only way to reduce one's uncertainty is through active engagement and one is literally applying active inference when they do actively engage in this epistemic way so I hope that all comes through all right a first question from spumato in the chat does active inference provide us with predictive power potentially focusing on human or social scale issues what does it look like or where would you say active inference provides us with predictive power or unique explanations and predictions so uh yes or no depending on the granularity of the question so what's your look at acting friends and not active in France action France is a branch of creative processing so it's a family in incognition that's consider that your perception of the world is based on prediction you have two specificity of active inference within practice processing one specificity is that it says that actions are predictive our base and prediction to the other is that it proposed basically a mathematical formalism in which to study that that grounds the claim but also constrain the mechanisms a little bit uh proactive processing as a family is predictive of um many things inspective of the fact that basically you think you see things but you are not in capacity to perceive them right now so I see bit of tissue no fabric that has been my computer but I know and I see the patterns in it but I know perfectly well that they sell that are actually that have the Precision to see those patterns they are looking at the screen now so I cannot possibly know what the buttons are except if I expect it and this is what checked my perception so this is predicted by predictive processing and what is also operated is that basically the information flow in the uh perception part of the brain it is heavily top down you have more uh bandwidth that goes from visor cortex to the retina then reciprocally which is simply not compatible with the direct perception uh passive perception kind of model it is however compatible with the idea that we predict the content of you know retail activity and when uh our prediction is wrong with then you have feedback you have a perception error propagation so this is pure practice processing this is explained by crazy processing this is contradictory actively with most conceivable variants of non-prex processing so yeah prex processing is uh productive and I say it's not up for debates as I know active in France it takes in addition the idea that um the actions are based on prediction I do not know what work empirical work existed to check whether this is specifically the case maybe Daniel you know what has been done about this complete your answer and I'll give a thought okay so there are considerable empirical settings that would check that out I do not know of them so I will not claim that they exist or do not

exist uh what is important to me is the mathematical grounding because uh most model of cognition kind of assume that we have computational power that is implemented somehow somewhere by you know means and this is wrong we cannot just suppose that the big properties that the brain have you know they must exist somehow somewhere we have to actually explain it and this is what the mathematical formalism behind active inference principle does it explains that we should expect a neural system to be predictive somehow and also it is not directly limited in math it is pretty easy to see why the brain should have a lot of power doing that because of the you know very heavy tune power for demicola tune monitors so but I would not if active inference is empirically distinct from other case of practice processing I am not sure how I cannot think of an experiment or empirical observation that definitely discriminates between Latin France and none 13 France brexit processing these things it exists great thank you well in many ways active inference is predictive processing with action under additional sophistication and richness supported by the free energy principle these are technical topics that we explore in other discussions as to whether active inference gives us unique explanation or prediction capacity one could look to for example the linear regression or any other type of statistical framework linear regressions in and of themselves don't make predictions because you need a Target to make a prediction about however given a linear regression model relating to health outcomes for example you might use that model to reduce your uncertainty which is to say have predictive power about that system of interest and so in that respect active inference has been used to develop a host of unique explanations and predictions related to a variety of cognitive and behavioral phenomena I'll just leave it at that and I'll read some introductions and such in the chat so Robin writes hi my name is Robin I work in digital governance yeah go ahead um what makes that I buy it in France as a mechanical model of human cognition is that you don't really get uh adaptive chemical achievement adaptive action without some variant of action being predictive because the default picture would be that you have to planify a full action trajectory but there is no computable way this can be fast enough and robust enough and not costly enough to underline adaptive action so you'd be basically you'd say I have to grab a cup I would look at the cup see the cup plentify movements another movements if something happens if I was slightly wrong about the position of the cup uh uh the algorithm is too heavy for me to adjust so you'll not be able to navigate anything ultimate information or this is changing and we are very good at navigating things that means inflation which are changing so until there is another mechanical model that predicts you know attuned adaptive action I

will have to buy active inference as a model of human conditions basically because the mass and the line is pretty uh solids in my opinion yeah it's always a great question is relative to what measurements need a comparator and so active inference with predictive value relative to something unspecified doesn't really make sense but we can evaluate it better or worse in a certain setting and so it's important to compare it so I'll just read a couple introductions Robin wrote hi my name is Robin I work in digital governance I've been curious about applying active inference to the construction of governance for digital Commons great kickoff swamato wrote hi I've been studying fundamental physics and now I am doing Clio Dynamics with a current interest in pragmatist philosophy I came to discover active inference and see how to apply it to cultural evolution under says hello Anthony writes bonjour Anthony writes background poker player father entrepreneur very little knowledge on subject attend cognitive Security in Davis working on project that active inference is very relevant to expect to learn and engage and Gina writes hi I'm a biologist and neuroscientist that wants to understand the theory to do experiments with the Xscape team cool well people can continue to add um introductions or or questions in the chat Ben if you want to um ask or or reflect on anything or I'm sure there's more things to talk about yeah I thought that was a fantastic introduction some very interesting connections that I wasn't expecting there um nice to see manufacturing consent making an appearance and a lot that I think I can uh hold on let me turn on my camera yeah certainly a lot that I uh want to pick up on and hopefully um Circle background to in the in the in the session that I'll be doing cool cool by the way what I wanted to say earlier and I could check now is that I believe I believe that manufacturing consent was a Shamsky thing but it is not it is a Litman thing which is the guy that basically funded the the very notion of public opinion in the 30s in the book called public opinion the book is co-authored with Chomsky learner Chomsky was not it was maybe eight at the time I very much doubt he actually did a thing so I'm sure you have a copy of that book on my shelf and I'm pretty sure of the book manufacturing concerned by Chomsky the expression manufacturing consent is uh from a 30s book public opinion by Walter McMahon that for me the whole public opinion thing according to Wikipedia they studied not uh well looked for a specific book that would specifically will be less public opinion because like what do we mean by public opinion I don't know Lord don't go clearly but maybe something else goes I don't know it's created for that I can at least say it that's kind of the interesting question about multi-scale systems and emergence if we enable or grant that brains can perceive or do things that a single neuron can't but that they have a material relationship to one another then is public opinion just like a pile

of sand it just represents the summation of the The Nest mates or does public opinion represent a singular integrated unit in its own respect one that might have to be sampled by investigating the material constituents but still being something different from that I'll just ask go to a question in the chat or apple if you ever thought of sand have semantics I adopted for an observerance or for itself sorry does the pile of sand have semantics for an observer or for itself good question foreign but I can put together brain and and public opinion different coordination mechanisms but still connected systems sense if the sand is piling up at the bottom of an hourglass it certainly has semantics for an observer who knows about hourglasses Michael says hello haptivist greetings Michael and asks a question please further contrast the terms regimes of attention and socio-cultural landscapes contrast as in explaining what the terms are I guess yeah how are those terms similar different they say I do not see uh basically we uh we I propose we active Influence People I did not have the idea uh propose regimes of attention so specific patterns in the way you you know your attention flows as an explanatory mechanism let us say what underlies the experience of cognitive landscape coincidence whatever is the array of things that he was an agent um experience uh the world and then uh we activated from people also claim that the coordination between individuals in uh you know cultural agency so the way we co-construct synchronize regime of attention means that there is a coupling between the landscape that we experience so landscape that are constrained in such a way I will call social cultural landscape because it's easier than saying cognitive landscape that are constrained by social cultural processes is that satisfying yeah it's great I'll just give another view on that a socio-cultural landscape is something like the ecological landscape or the actual basis materially of our system of Interest what we are studying in sociology and so it consists of inert objects as well as animate cognitive entities whereas regimes of attention refers to the dynamical patterns of attention which is to say relevance and salience for cognitive agents and so the socio-cultural landscape to use an example from avils presentation might include the modified Niche that includes a monument and then a given cognitive agents might when they gaze upon that landscape in a in cultured fashion have a regime of attention such that their eyes are moved towards the point of a monument but the regime of attention is modeled as a pattern of dynamics of a single cognitive Asians and inactive inference we model attention as covert or internal I want to add something I use possibly abuse the term landscape because it has a very specific meaning slash connotation which is a space that you can gaze at that this a landscape is a space that we can gaze at and you if you go to a more mathematical notion of space you will basically find

very closely the expectation or formal entitlements but this space represents the set of possibilities accessible to a system like that this is all there is to and this is not what the cognitive landscape is it's not the set of possibilities that abstractly an agent could experience it is what an agent experience is no what it can they can guys at no um so this is why the notion of a landscape is very heavily used it does not have a clear mathematical meaning it should have because it has a very specific one conceptual meaning that is critical to the understanding we have of cognition and that is not afforded by um set theory so most mathematics we would use um the brand of mathematics we teach in school and the physicists use most physicists not all physicists although we consider like waddington's epigenetic landscape or if we consider the surface that variational free energy is optimized on as a landscape it gives us a lot of really good intuitions it gives us something to gaze upon as instrumental modelers but it is not the same as the landscape of trees even though each one of those trees might be modeled by an observer as possessing a epigenetic landscape or a regime of attention or a free energy landscape well if anyone has other thoughts or questions we can consider them otherwise uh just give me a signums I don't think and everyone has a H do you see the uh the image yeah it's small but waddington's landscape there you go thank you yeah Wellington's landscape so it is a very old uh 30s 20s 30s I guess thing in um evolutionary biology where so Weddington is let us say the founder of the Contemporary developmental biology and the landscape are basically a dynamical Fitness I do not know how to not sail landscape that is a problem there are basically the space in which uh agents can evolve the phenotype of the agent can evolve during its development um this is a very very very um Grande visual metaphor symbolism I don't know for the story of Developmental system so this is why we told koflandscape for the gazing and for this historical grounding of the developmental studies in biology and also I literally I'm paid to landscape biology landscape archeology so yeah let's overuse it on landscape that is good for me cool but it's an example of using our spatial and embodied experience to to get a grasp on different conceptual topics Ben you have any comments or or I'd love to hear a little yeah actually ask about this this image if there's a couple of words that I was expecting to creep into this discussion that that did not which are uh attractors and repellers about when you mentioned dynamical systems I wonder if you think these are useful terms for thinking about because attract attraction goes very naturally with thinking about attention and I wonder if you think this is a useful um kind of linguistic tool for visualizing the relationship between uh social cognitive Landscapes and uh and attention um do you see these children trains diagram now okay I wanted to add it if there

is a screenshot laying around I wanted this to be a to ensure so attractors and repellers yeah I mean with basically talking Fitness when we're talking landscape Fitness Landscapes so I guess it kind of entails the notion of attractors and repellers uh no there are many reasons why it stay away from the mikola systems terminology because well chemical systems I can tell the preformulation of all possibilities of the system which I think is very very very wrong to do for biological system and social system that are that undergo uh the evolution and reconstruction so I stay away from this uh formalism whenever possible and attractor technically mean sub-regions that you cannot possibly get out which exists in conservative system basically but uh in all cases that we're interested about it's technically not a tractor um we can't think of it as a structure at a specific time scale it's okay but then if we have to uh abuse the formalism this much why not not use the formalism that would be my uh flow of reflection about this terminology so I understand the tractors as uh states that have visited more regularly than others so is there something about that that's not correct I will check it out yeah an attracting set is one that is Revisited but there are of course many sub definitions to be more technical like the attractor is it just the point or is it the Basin of Attraction but broadly on a landscape as things shake out the attractor begets some states that are Revisited more than those that are not that's kind of the tautological definition of the attracting set so so uh a director is emphatically not at least under a standard definition uh set of states that is visited more often it is characterized by forward invariance so the fact that once you're in you don't get out so an attractor is a manifold so a Subspace let us say such that if you're in get out well there is a busting of Attraction such that if you get in the Basin you get into the so under if you look at stochastic clinical systems maybe you will get out because it's you know a stochastic means with random noise basically let's say that so if you are specifically within a stochastic technical system so an optimical system per c um then you can talk of attractor and think of things that are not actually uh that do not have this forward invariance just things that you tend to stay in for a while once you win but you are not talking about contractor anymore you are talking about some other thing the functionality is foreign we shouldn't fall down this Rabbit Hole completely we can talk about this another time let me go to another great question in the chat Michael says a philosophical question science does a beautiful job of describing and building testable predictions of reality but underweight subjective experience human subjectivity arises within experience of observation but includes preconditioning as well as experience how should science informed observers reconcile the limits of subjective experience with the limits of objectification science build good models good models are a

competent engagement that is basically it um so the question of how to build a sense that is effective to information Kairos I will not use the word effective perceive but you know that's pretty close and we'd have to in principle sense is built to uh create precise models precise adding does not create prediction error within a specific context that is a um heavily heavily ritualistic and conservative and that would be you could criticize pretty heavily the institution of science as a truth Seeker normative but so you you'd have to have a good level of Reform to basically enforce a modeling that are both robust and easy to transmit socially outside the cast of scientists so uh Danielle and nine many other people wrote a paper on ontology for decentralized science which speaks to it and um yeah so if you get science you get good models and once you are good models you can navigate competently but slash and I don't know but and both what then then uh band um you would need a heavy level of Reform institution of science to get this process be more you know ergonomic efficient compelling and socially influential and uh yeah I think it's an important thing to do when I try to do it so thank you for the question I'll give another angle on that Bayesian statistics qualitatively equates to the explicit provisioning of our priors and how we update those prior beliefs as new observations come in we may have high attention to new observations and update our beliefs a lot we may not be paying to the attention to the new stimuli and therefore continue our prior belief so if we're not paying attention it's kind of like a belief with inertia that's greater and if we are paying a lot of attention then beliefs change more rapidly and trying to capture every single belief that leads to a research team whether they're in Academia or not to carry out interpret analyze and communicate a certain topic that is a really far-reaching task however Bayesian statistics and Bayesian approaches like active inference are helping us get a grasp at the preconditioning that does play an essential role with ongoing experience the question of how to engage with extra institutions and norms and practices is a challenging and ongoing work but in some ways the formal principles that help us understand how context and measurement come together for cognitive entities and groups those fundamental principles are being communicated maybe also I'd love to hear about what you're planning to talk about yeah I'm not sure if I if I uh I'm not sure if I even fully understand the question I took it the question just briefly I took it to be maybe some worry about the the relationship between the formalisms and computational models in active inference and human subjectivity itself so how do they how do they bridge the gap of phenomenology there it's a really interesting question I know that there are some people who think that phenomenology is human subjective experience is just going to have these transcendental aspects to it that can't be captured um

with by the formalisms um that's something I won't be talking about in my session what I will be talking about is um so some of the so I'm going to speak on some things that did come up briefly today um so many of the readings uh that feature in my section are uh readings that will use the term predictive processing so I was going to say a little bit quite briefly about how I think these different um different slightly different Frameworks fit together and what some of the similarities and differences are I was going to say move quite quickly and superficially over some of the the core mechanisms so Precision waiting predictions generative models um and then I was the what I'm going to try and do is before we go on to the subsequent weeks where we look at Collective behavior um Norms normativity social scripts and things I wanted to try and build up a nicely layered picture of what it's like to be an active inference agent in the world and so the way I'm going to do this is by um swooping over some philosophical territory so I'm going to look at the way active inference has um said interesting things about philosophy of Mind particularly in terms of the role of internal representation um if I have time I'm going to say a little bit about embodied cognition some of the four re stuff um Avo you got it right so it's embedded embodied enacted and extended and I'm not going to be able to talk about all of those but I'm going to say a little bit about some of them and how active inference kind of brings interest in debates in that area out I'm going to say a little bit about folk psychology and motivated action so talking briefly about what agency itself is for an active inference agent um and whether it's psychology folk psychology it's um it's a kind of propositional psychology so it's a it's a framework of propositions that we generally use to describe and explain human behavior so it involves uh propositions such as a desires b or X hopes that Y and it turns out that this propositional psychology is pretty good um for explaining and predicting human behavior but it's not just say it's a scientistly persons philosophers so do you mean we in the sense that it's it's a useful framework um I mean that you should explain what the term is because uh it's full psychology is quite specific it's not a word that is in you know folk psychology oh okay okay yeah so folk psychology is just that basically the linguistic framework we use to describe human behavior so if I say you know if I want to explain why um aval goes to the refrigerator to grab a beer I would say you believe there's beer in the fridge you desire a beer and then on the basis of that you form the intention to then go to the fridge um and it's been the subject of a lot of philosophical interest because the the kind of ontological status of those Concepts is very unclear so it's it's not clear whether when we use folk psychological concepts are we saying that those those Concepts exist in some meaningful way or are we taking them more or should we take a more pragmatic

approach and say that this is just instrumentally useful but we're not actually positing the existence of any kind of brain States or anything like that um and there's been some interesting work on how predicted processing and active inference might map onto folk psychology um are they in a necessary kind of tension because obviously with predictive processing you have um just a very very limited set of Primitives with which to describe and explain human behavior so I'm going to say a little bit about um and then I'm going to move straight into talking about phenomenology and talking about how um certain aspects of the framework have been said to underpin certain kinds of experiences so I'm going to use a case study of addiction I think possibly depression but I think I'm going to talk about addiction and with this case study I'm hoping to bring out um kind of tie together all of the things that I've said up to that point and some of the things that you've covered as well and I should say as well in the discussion session um for my section we'll have a special guest as well so we're going to be joined by Professor Mark Miller who is um a prolific author of predictive process in a neurophenomenology type stuff he's done some really really interesting work on happiness and well-being depression addiction and really using the active inference framework in some quite imaginative ways um and he's awesome in discussion so um hopefully you'll find the readings for my section really interesting and if you do you'll have a chance to um kind of have a conversation with the author for those pieces travel any closing notes actually influence rules I will say one short question I just want to get it on there while we're here Bryn asks as a psychologist I am more or less in the business of helping people minimize their variational free energy and expected free energy great profession but what are the markers of free energy in the brain or organism is it heart rate neuromodulation galvanic skin response cortisol levels Etc I say the uh closer proxy would be uh so those prediction error things in the brain so when we have expectation that our related to other preparation of these information the fact the expectation was variated and this would be the closest uh proxy to rational free energy I do not think it would be the exclusive one but it is the main one and you could I don't think you could make an exhaustive list basically anything that is out of order so as to prompt any kind of response of either you know updating models or facting on things it would be a marker of uh well prediction error which is not quite version for energy but is also quite not not quite not it's so far expected free energy um I have no idea whatsoever I don't think this framework is uh has made uh it's connected explicitly to physiology yet Am I Wrong Daniel yeah it's very uh it's very interesting question and all I will say or all I feel like I can say on it is um one thing we're going to talk about in my session is um how prediction error is not always a bad thing so

there's a really nice aeon article that came out fairly recently about the value of uncertainty and one of the things I'm going to talk about is how um predictive processing has you know perhaps it can account for phenomenons such as uh playful Behavior creativity um engagement in challenging hobbies and dangerous sports and things like this and how we get an account of exploratory behavior and curiosity which is kind of rooted in the fact that for for creatures like we are for active inference agents um the resolution of prediction error itself is a good thing and so in some sense we're always seeking it out so it's I think it might be a mistake to kind of in it's very intuitive to equate prediction error or free energy with kind of allostatic load and stress and negativity but actually it's going to turn out that that's not the case yes thank you and I'll give another type of answer which is variational free energy and expected free energy our statistical calculations about statistical models so bodies don't have expected free energies models that are made about bodies can have variational and expected free energy and so to that end whatever you put in your model is either an observable or a proposed latent hidden state that you didn't measure and you may find that different experiential reports or different physiological phenomena have different relationships to each other so indeed it has been explorers for example which neural firing rates are associated with surprise or which hormonal changes are associated with this or that change in body temperature when the generative model is made the deflationary answer is look at the data and find out which cues you measure are associated with your calculation of vfv and efe however I think it's a little bit of a map territory situation and one that's going to come up I am sure many times in our course great well I'll just remind people to go to the course website to review the syllabus and also encourage everybody to submit questions through the course website all of the teachers will review the questions and we'll give the best answers we can and also please from all backgrounds we would love to hear your voice in the discussions so thank you again Ben avail see you next week bye